

RASITOL[®] TABLET 40MG

Ingredient(s):

Each tablet contains:

Furosemide 40mg

Pharmacodynamics:

1. Furosemide is one of the most potent diuretics. It generally produces an adequate increase in sodium and water excretion even in cases in which diuresis cannot be increased in other ways.
2. Furosemide inhibits reabsorption of sodium and water in the ascending limb of the loop of henle by interfering with the chloride binding site of the Na^+ , K^+ , 2CL^- -cotransport system.
3. Furosemide increases the rate of delivery of tubular fluid and electrolytes to the distal sites of hydrogen and potassium ion secretion, while plasma volume contraction increases aldosterone production. This increased delivery and high aldosterone levels promote sodium reabsorption at the distal tubules, thus increasing the loss of potassium and hydrogen ions.
4. It is effective even in patients with considerably impaired glomerular filtration, hypoalbuminaemia or metabolic acidosis.
5. Furosemide lowers blood pressure initially by reducing plasma and extracellular fluid volume; cardiac output also decreases. Eventually, cardiac output returns to normal with an accompanying decrease in peripheral resistance.

Pharmacokinetics:

1. Furosemide is fairly rapidly absorbed from the gastro-intestinal tract; bioavailability has been reported to be about 60~70% by absorption is variable and erratic.
2. The half-life of furosemide is up to about 2 hours although it is prolonged in neonates and in patients with renal and hepatic insufficiency. It is up to 99% bound to plasma albumin, and is mainly excreted in the urine, largely unchanged. There is also some excretion via the bile and non-renal elimination is considerably increased in renal impairment.
3. Furosemide crosses the placental barrier and is distributed into breast milk. The clearance of furosemide is not increased by haemodialysis.

Indication(s):

Oedema (cardiac, hepatic, renal) and hypertension.

Dosage & Administration:

- Adult: For oedema, the usual initial dose is 1 tablet once daily, adjusted as necessary according to response.
For hypertension, a daily doses of 1-2 tablets is given, either alone, or in conjunction with other antihypertensives.
- Elderly: In the elderly furosemide is generally eliminated more slowly. Dosage should be titrated until the required response is achieved.

Route of Administration: Oral

Contraindication(s):

1. Furosemide should not be used in case of renal failure with reduced or absent urinary excretion (oliguria anuria) resulting from poisoning with nephrotoxic or hepatotoxic agents. This also applies to severely impaired renal function with simultaneous liver failure and impaired consciousness (hepatic coma).
2. Severe hypokalaemia must be compensated before furosemide treatment. Potassium loss may be due to unbalanced diet, to vomiting or diarrhoea or to diseases of the liver or adrenal cortex and, in diabetics, to hyperglycaemia.
3. Severe electrolyte depletion (e.g. potassium or sodium) as well as serious reduction in blood volume (hypovolaemia with or without hypertension) and hypotension must be compensated before furosemide is administered.
4. Furosemide is contraindicated with patients who have a history of hypersensitivity to it as well as in women with childbearing potential.

Side Effect(s) / Adverse Reaction(s):

1. The most common side-effect associated with furosemide therapy is fluid and electrolyte imbalance including hyponatraemia, hypokalaemia, and hypochloraemic alkalosis, particularly after large doses or prolonged administration. Because of its shorter duration of action, the risk of hypokalaemia may be less than thiazide diuretics.
2. Unlike the thiazide diuretics, furosemide increases the urinary excretion of calcium. Nephrocalcinosis has been reported when furosemide has been used to treat preterm infants.
3. Furosemide may provoke hyperglycaemia and glycosuria, but probably to a lesser extent than the thiazide diuretics. It may cause hyperuricaemia and precipitate attacks of gout in some patients.
4. Signs of electrolyte imbalance include headache, hypotension, muscle cramps, dry mouth, thirst, weakness, lethargy, drowsiness, restlessness, oliguria, cardiac arrhythmias, and gastro-intestinal disturbances.
5. Other side effects are relatively uncommon, and include blurred vision, yellow vision, dizziness, headache and orthostatic hypotension. Skin rashes, photosensitivity reaction and hypersensitivity reactions including interstitial nephritis, vasculitis, and fever occur rarely.
6. Pancreatitis is more common at high doses and cholestatic jaundice has been reported. Bone marrow depression may occur rarely: agranulocytosis, thrombocytopenia, and leucopenia have been reported.

Precaution(s) / Warning(s) :

1. Although furosemide is used in high doses for oliguria due to chronic or acute renal insufficiency it should not be given in anuria or in renal failure due to nephrotoxic or hepatotoxic drugs nor in renal failure associated with hepatic coma.
2. Furosemide should not be given in pre-comatose states associated with hepatic cirrhosis.
3. It should be used with care in patients with prostatic hyperplasia or impairment of micturition since it can precipitate acute urinary retention.
4. Pregnancy and lactating mother: Furosemide crosses the placenta. Furthermore, it may diminish the utero-placenta blood flow. Therefore, administration of furosemide during pregnancy should be avoided if there is no strict indication for the use of it.

Furosemide also passes into breast milk, thus patients treated with it should not breast-feed. It also inhibits lactation.

5. Patient with manifest or latent diabetes mellitus should be checked regularly during furosemide treatment. The same applies to patients with gout.

Drug interaction(s):

1. Furosemide may increase the ototoxic potential of aminoglycoside antibiotics, especially in the presence of impaired renal function. Except in life-threatening situations, avoid this combination.
2. Lithium generally should not be given with diuretics because they reduce lithium's renal clearance and add a high risk of lithium toxicity.
3. Furosemide may add to or potentiate the therapeutic effect of other antihypertensive drugs. Potentiation occurs with ganglionic or peripheral adrenergic blocking drugs.
4. Furosemide may enhance the nephrotoxicity of cephalosporin antibacterials (e.g. cephalothin).
5. The diuretic effect of furosemide has been shown to be substantially reduced by concomitant antiepileptic therapy (i.e. carbamazepine and phenobarbitone).
6. NSAIDs may antagonise the diuretic effect of furosemide and other diuretics. Concomitant administration of NSAIDs with diuretics may increase the risk of nephrotoxicity.
7. Probenecid has been shown to reduce the renal clearance of furosemide and reduce the diuretic effect.
8. When a cardiac glycoside is administered concurrently, it should be remembered that potassium deficiency increases the sensitivity of the myocardium to digitalis.
9. Furosemide sometimes diminishes the potency of other drugs (e.g. the effect of antidiabetics and pressor amines) or potentiates their effect (e.g. in the case of salicylates, lithium and curaremimetic muscle relaxants). The action of other hypotensive drugs may be potentiated by furosemide.

Symptoms and Treatment of Overdose:

The principle signs and symptoms of overdose with furosemide are dehydration, blood volume reduction, hypotension, electrolyte imbalance, hypokalemia and hypochloremic alkalosis, and are extensions of its diuretic action.

Treatment of overdosage is supportive and consists of replacement of excessive fluid electrolyte losses. Serum electrolytes, carbon dioxide level and blood pressure should be determined frequently.

Shelf Life:

3 Years from the date of manufacture.

Storage Condition:

Store at temperature below 30°C. Protect from light and moisture.

Product Description and Packing(s):

A white color elliptical tablet, one side impressed with a score. “”.

Blister packing of 10's x 10 and 10's x 100.



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